

Generative Pre-trained Transformer (GPT) ①

$$E \in \mathbb{R}^{V \times d} \quad x_i = E_{*i} \quad X \in \mathbb{R}^{T \times d}$$

Trainable Positional Encoding $P \in \mathbb{R}^{T \times d}$

Number of head $H \rightarrow d_h = \frac{d}{H}$

Input of Block l $X^{(l-1)} \rightarrow X^{(0)} = X + P$

$$W_Q^h, W_K^h, W_V^h \in \mathbb{R}^{d \times d_h}$$

$$Q_h = X^{(0)} W_Q^h$$

$T \times d_h \quad T \times d \quad d \times d_h$

$$K_h = X^{(0)} W_K^h$$

$T \times d_h \quad T \times d \quad d \times d_h$

$$V_h = X^{(0)} W_V^h$$

$T \times d_h \quad T \times d \quad d \times d_h$

$$S_h = \frac{Q_h K_h^T}{\sqrt{d_h}}$$

$T \times T \quad T \times d_h \quad d_h \times T$

$$\tilde{S}_h = S_h + M$$

$T \times T \quad T \times T \quad T \times T$

where $M_{ij} = \begin{cases} 0 & j \leq i \\ -\infty & j > i \end{cases}$

$$A_h = \text{softmax}_{\text{row}}(\tilde{S}_h)$$

$T \times T \quad T \times T$

if $Z = [z_1, z_2, \dots, z_T]$

$$\text{softmax}_{\text{row}}(Z_i) = \frac{Z_i}{\sum_{j=1}^T e^{Z_j}}$$

$$O_h = A_h V_h$$

$T \times d_h \quad T \times T \quad T \times d_h$

$$O = \text{concat}(O_1, O_2, \dots, O_H) W_O$$

$T \times d \quad T \times d \quad d \times d$

$$W_O \in \mathbb{R}^{d \times d}$$

Residual connection

$$R = X^{(0)} + O$$

$T \times d \quad T \times d \quad T \times d$

$R \in \mathbb{R}^{T \times d}$

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$$Y = \text{LayerNorm}(R)$$

 $T \times d$ $T \times d$

The LayerNorm operation is defined as follows:

$$Z = \frac{X - \mu}{\sigma}$$

$$\mu = \frac{1}{d} R \mathbf{1}_d$$

where $\mathbf{1}_d \in \mathbb{R}^{d \times 1}$, $\mu \in \mathbb{R}^{T \times 1}$

$$\therefore R \mathbf{1}_d = \begin{bmatrix} R_{11}, \dots, R_{1d} \\ R_{21}, \dots, R_{2d} \\ \vdots \\ R_{T1}, \dots, R_{Td} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} R_{11} + R_{12} + R_{13} + \dots + R_{1d} \\ R_{21} + R_{22} + R_{23} + \dots + R_{2d} \\ \vdots \\ R_{T1} + R_{T2} + R_{T3} + \dots + R_{Td} \end{bmatrix}$$

$$\mu \mathbf{1}_d^T$$

 $T \times d$

$$\begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_T \end{bmatrix} \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} = \begin{bmatrix} \mu_1 & \mu_1 & \dots & \mu_1 \\ \mu_2 & \mu_2 & \dots & \mu_2 \\ \vdots & \vdots & \dots & \vdots \\ \mu_T & \mu_T & \dots & \mu_T \end{bmatrix}$$

$$C = R - \mu \mathbf{1}_d^T$$

 $T \times d$ $T \times d$ $T \times d$

where $C \in \mathbb{R}^{T \times d}$

$$C \odot C$$

 $T \times d$ $T \times d$

means $(C \odot C)_{ij} = C_{ij}^2$

$$\sigma^2 = \frac{1}{d} (C \odot C) \mathbf{1}_d$$

 $T \times 1$ $T \times d$ $d \times 1$

where $\sigma^2 \in \mathbb{R}^{T \times 1}$

$$\begin{bmatrix} C_{11}^2 & C_{12}^2 & \dots & C_{1d}^2 \\ C_{21}^2 & C_{22}^2 & \dots & C_{2d}^2 \\ \vdots & \vdots & \dots & \vdots \\ C_{T1}^2 & C_{T2}^2 & \dots & C_{Td}^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} C_{11}^2 + C_{12}^2 + \dots + C_{1d}^2 \\ C_{21}^2 + C_{22}^2 + \dots + C_{2d}^2 \\ \vdots \\ C_{T1}^2 + C_{T2}^2 + \dots + C_{Td}^2 \end{bmatrix}$$

$$D = \sqrt{\sigma^2 + \epsilon} \mathbf{1}_T$$

 $T \times 1$ $T \times 1$ $T \times 1$

where $\mathbf{1}_T \in \mathbb{R}^{T \times 1}$, $\epsilon = 10^{-5}$, $D \in \mathbb{R}^{T \times 1}$

ϵ is used in order to avoid division by 0

$$\hat{R} = C \odot (D 1_d^T)$$

where $\hat{R} \in \mathbb{R}^{T \times d}$ row wise broadcast

$$\begin{bmatrix} C_{11} & C_{12} & \dots & C_{1d} \\ C_{21} & C_{22} & \dots & C_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ C_{T1} & C_{T2} & \dots & C_{Td} \end{bmatrix} \odot$$

$$\begin{bmatrix} D_1 \\ D_2 \\ \vdots \\ D_T \end{bmatrix} [1 \ 1 \ \dots \ 1]$$

$$\hat{R} = C \odot (D^{-1} 1_d^T)$$

$$\hat{R} \odot D 1_d^T = [C \odot (D^{-1} 1_d^T)] \odot D 1_d^T$$

$$C = \hat{R} \odot D 1_d^T$$

$$\begin{bmatrix} D_1 & D_1 & \dots & D_1 \\ D_2 & D_2 & \dots & D_2 \\ \vdots & \vdots & \ddots & \vdots \\ D_T & D_T & \dots & D_T \end{bmatrix}$$

$$\Gamma = 1_T \gamma$$

where $1_T \in \mathbb{R}^{T \times 1}$, $\gamma \in \mathbb{R}^{1 \times d}$, $\Gamma \in \mathbb{R}^{T \times d}$

broadcast across all rows

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} [\gamma_1 \ \gamma_2 \ \dots \ \gamma_d] = \begin{bmatrix} \gamma_1 & \gamma_2 & \dots & \gamma_d \\ \gamma_1 & \gamma_2 & \dots & \gamma_d \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_1 & \gamma_2 & \dots & \gamma_d \end{bmatrix}$$

$$B = 1_T \beta$$

where $1_T \in \mathbb{R}^{T \times 1}$, $\beta \in \mathbb{R}^{1 \times d}$, $B \in \mathbb{R}^{T \times d}$

$$Y = \hat{R} \odot \Gamma + B$$

where $Y \in \mathbb{R}^{T \times d}$

LayerNorm ends!

$$U = Y W_1 + 1_T b_1$$

$T \times d_{\text{ff}}$ $T \times d_{\text{ff}} \times d_{\text{ff}}$

where $W_1 \in \mathbb{R}^{d \times d_{\text{ff}}}$, $b_1 \in \mathbb{R}^{1 \times d_{\text{ff}}}$, $U \in \mathbb{R}^{T \times d_{\text{ff}}}$

$$d_{\text{ff}} \approx 4d, \ 1_T b_1 = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} [b_1 \ b_2 \ \dots \ b_{d_{\text{ff}}}]$$

$$G = \text{softmax}(U)$$

where $G \in \mathbb{R}^{T \times d_{\text{ff}}}$

$$= \begin{bmatrix} b_1 & b_2 & \dots & b_{d_{\text{ff}}} \\ b_1 & b_2 & \dots & b_{d_{\text{ff}}} \\ \vdots & \vdots & \ddots & \vdots \\ b_1 & b_2 & \dots & b_{d_{\text{ff}}} \end{bmatrix}$$

$$F = G W_2 + 1_T b_2$$

where $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{ff}}}$, $b_2 \in \mathbb{R}^{1 \times d_{\text{ff}}}$

$$\boxed{G = \text{GELU}(U)} \quad \begin{matrix} T \times d_{ff} & T \times d_{ff} \end{matrix}$$

where

$$\text{GELU}(x) = x \Phi(x) \text{ or } \text{GELU}(x)$$

$$G \in \mathbb{R}^{T \times d_{ff}}$$

$$\approx \frac{x}{2} \left[1 + \tanh\left(\frac{\sqrt{2}}{\sqrt{\pi}} (x + 0.044715x^3)\right) \right]$$

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$$\boxed{F = G W_2 + \mathbf{1}_T b_2}$$

$$\begin{matrix} T \times d & T \times d_{ff} & d_{ff} \times d & T \times d \end{matrix}$$

where $W_2 \in \mathbb{R}^{d_{ff} \times d}$, $b_2 \in \mathbb{R}^{1 \times d}$

$$\mathbf{1}_T b_2 = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} [b_1 \ b_2 \ b_3 \ \dots \ b_d]$$

$$= \begin{bmatrix} b_1 & b_2 & b_3 & \dots & b_d \\ b_1 & b_2 & b_3 & \dots & b_d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_1 & b_2 & b_3 & \dots & b_d \end{bmatrix}$$

$T \times d$

$$\boxed{Z = Y + F}$$

$$\begin{matrix} T \times d & T \times d & T \times d \end{matrix}$$

where $Z \in \mathbb{R}^{T \times d}$

$$\boxed{X_1 = \text{LayerNorm}(Z)}$$

The same LayerNorm operation described earlier.

The operations described so far constitute one Transformer decoder block. The resulting (X) is then fed into the next decoder block, where the same operations are repeated using different weights and biases.

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$X^{(n)} \in \mathbb{R}^{T \times d}$ the output of the last Transformer decoder block

$L \in \mathbb{R}^{T \times V}$ L is Language Model Head (LM Head)

V is the number of words (Vocabulary)

$$L = X^{(n)} W_{LM} \quad \text{where } W_{LM} \in \mathbb{R}^{d \times V}, \text{ use } E^T \text{ as } W_{LM}$$

$T \times V \quad T \times d \quad d \times V$

in order to share word representations between the input and output.

$$P = \text{softmax}(L)$$

$T \times V \quad T \times V$

$$\mathcal{L} = -\frac{1}{T} \sum_i \log P_{i, y_i}$$

where i is the token position in the sequence.
 y_i is the vocabulary index of the correct target token at sequence position i .

Forward propagation ends!

Back propagation of $P = \text{softmax}_{\text{row}}(L)$ and

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$$L = -\frac{1}{T} \sum_{i=1}^T \log P_{i, y_i}$$

$$L = -\frac{1}{T} \sum_{i=1}^T \sum_{j=1}^V Q_{ij} \log P_{ij} \quad \text{where } Q_{ij} \text{ is one hot encoding}$$

$$dL = -\frac{1}{T} \sum_{i,j} \frac{Q_{ij}}{P_{ij}} dP_{ij} \quad P, Q \in \mathbb{R}^{T \times V}$$

$$\frac{\partial P_{ij}}{\partial L_{ik}} = P_{ij} (\delta_{jk} - P_{ik}) \quad \text{where } \delta_{jk} = \begin{cases} 1 & j=k \\ 0 & j \neq k \end{cases}$$

$$dP_{ij} = \sum_{k=1}^V \frac{\partial P_{ij}}{\partial L_{ik}} dL_{ik}$$

$$dP_{ij} = \sum_{k=1}^V P_{ij} (\delta_{jk} - P_{ik}) dL_{ik}$$

$$dL = -\frac{1}{T} \sum_{i,j} \frac{Q_{ij}}{P_{ij}} \sum_k P_{ij} (\delta_{jk} - P_{ik}) dL_{ik}$$

$$= -\frac{1}{T} \sum_{i,j,k} Q_{ij} (\delta_{jk} - P_{ik}) dL_{ik}$$

$$= \sum_{i,k} \left(-\frac{1}{T} \sum_{j=1}^V Q_{ij} (\delta_{jk} - P_{ik}) \right) dL_{ik}$$

$$-\frac{1}{T} \sum_{j=1}^V Q_{ij} (\delta_{jk} - P_{ik}) = -\frac{1}{T} \left[\sum_j Q_{ij} \delta_{jk} - \sum_j Q_{ij} P_{ik} \right]$$

where $\sum_j Q_{ij} \delta_{jk} = Q_{ik}$, $\sum_j Q_{ij} P_{ik} = P_{ik}$ because $\sum_j Q_{ij} = 1$

$$\therefore dL = \sum_{i,k} \left[-\frac{1}{T} (Q_{ik} - P_{ik}) \right] dL_{ik} = \sum_{i,k} \frac{1}{T} (P_{ik} - Q_{ik}) dL_{ik}$$

$$\therefore (G_L)_{ik} = \frac{1}{T} (P_{ik} - Q_{ik})$$

$$\therefore \boxed{G_L = \frac{1}{T} (P - Q)} \quad \text{where } G_L \in \mathbb{R}^{T \times V}$$

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$$L = X^{(n)} W_{LM} \quad \text{Backpropagation}$$

$$\frac{\partial L}{\partial X^{(n)}} = \frac{\partial L}{\partial X^{(n)}} W_{LM}$$

$$\boxed{\frac{\partial L}{\partial X^{(n)}} = \frac{\partial L}{\partial X^{(n)}} W_{LM}^T}$$

Txv Txv vxd

$$\boxed{\frac{\partial L}{\partial W_{LM}} = \left(\frac{\partial L}{\partial X^{(n)}} \right)^T \frac{\partial L}{\partial X^{(n)}}$$

dxv dxT Txv

$$\rightarrow \frac{\partial L}{\partial W_{LM}} = \frac{\partial L}{\partial E}^{(LM)} \rightarrow \frac{\partial L}{\partial E} = \frac{\partial L}{\partial E}^{(Embedding)} + \frac{\partial L}{\partial E}^{(LM)}$$

$$\boxed{\frac{\partial L}{\partial E}^{(LM)} = \frac{\partial L}{\partial X^{(n)}}^T X^{(n)}}$$

$$X^{(n)} = \text{LayerNorm}(Z) \quad \text{Backpropagation}$$

$$X_{Txv}^{(n)} = \sum_{Txv} \odot \Gamma_{Txv} + B_{Txv}$$

$$X_{ij}^{(n)} = \sum_{ij} \odot \Gamma_{ij} + B_{ij}$$

$$dX_{ij}^{(n)} = \Gamma_{ij} d\hat{Z}_{ij}$$

$$dL = \sum_{ij} \left(\frac{\partial L}{\partial X^{(n)}} \right)_{ij} dX_{ij}^{(n)}$$

$$dL = \sum_{ij} \left(\frac{\partial L}{\partial X^{(n)}} \right)_{ij} \Gamma_{ij} d\hat{Z}_{ij}$$

$$\therefore \left(\frac{\partial L}{\partial Z} \right)_{ij} = \left(\frac{\partial L}{\partial X^{(n)}} \right)_{ij} \Gamma_{ij} \rightarrow \boxed{\frac{\partial L}{\partial Z} = \frac{\partial L}{\partial X^{(n)}} \odot \Gamma}$$

Txv Txv Txv

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$$X_{ij}^{(w)} = \hat{Z}_{ij} \Gamma_{ij} + B_{ij}$$

$$dX_{ij}^{(w)} = \hat{Z}_{ij} d\Gamma_{ij}$$

$$\begin{aligned} d\mathcal{L} &= \sum_{ij} (G_X^{(w)})_{ij} dX_{ij}^{(w)} \\ &= \sum_{ij} (G_X^{(w)})_{ij} \hat{Z}_{ij} d\Gamma_{ij} \end{aligned}$$

$$\therefore (G_\Gamma)_{ij} = (G_X^{(w)})_{ij} \hat{Z}_{ij} \rightarrow \boxed{G_\Gamma = G_X^{(w)} \odot \hat{Z}}$$

$T \times d \quad T \times d \quad T \times d$

$$\Gamma = \mathbf{1}_T \mathbf{x}$$

$$d\Gamma = \mathbf{1}_T d\mathbf{x}$$

$$\begin{aligned} d\mathcal{L} &= \text{tr}(G_\Gamma^T d\Gamma) = \text{tr}(G_\Gamma^T \mathbf{1}_T d\mathbf{x}) = \text{tr}(G_\Gamma^T (\mathbf{1}_T)^T d\mathbf{x}) \\ &= \text{tr}[(\mathbf{1}_T)^T G_\Gamma^T d\mathbf{x}] \end{aligned}$$

$$\therefore \boxed{G_X = \mathbf{1}_T^T G_\Gamma}$$

$1 \times d \quad 1 \times T \quad T \times d$

$$X_{ij}^{(w)} = \hat{Z}_{ij} \Gamma_{ij} + B_{ij}$$

$$dX_{ij}^{(w)} = dB_{ij}$$

$$d\mathcal{L} = \sum_{ij} (G_X^{(w)})_{ij} dX_{ij}^{(w)} = \sum_{ij} (G_X^{(w)})_{ij} dB_{ij}$$

$$\therefore (G_B)_{ij} = (G_X^{(w)})_{ij} \rightarrow \boxed{G_B = G_X^{(w)}}$$

$T \times d \quad T \times d$

$$B = \mathbf{1}_T \beta \rightarrow dB = \mathbf{1}_T d\beta$$

$$\begin{aligned} d\mathcal{L} &= \text{tr}(G_B^T dB) = \text{tr}(G_B^T \mathbf{1}_T d\beta) = \text{tr}[G_B^T (\mathbf{1}_T)^T d\beta] \\ &= \text{tr}[(\mathbf{1}_T^T G_B^T) d\beta] \end{aligned}$$

$$\therefore \boxed{G_\beta = \mathbf{1}_T^T G_B}$$

$1 \times d \quad 1 \times T \quad T \times d$

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$$\hat{Z} = C \odot (D 1_d^T) \quad \text{Backpropagation}$$

$$\hat{Z}_{ij} = \frac{C_{ij}}{D_i}$$

$$d\hat{Z}_{ij} = \frac{1}{D_i} dC_{ij} - \frac{C_{ij}}{D_i^2} dD_i$$

Path 1 Path 2

Path 1

$$d\mathcal{L} = \sum_{ij} (G\hat{Z})_{ij} d\hat{Z}_{ij} = \sum_{ij} (G\hat{Z})_{ij} \frac{1}{D_i} dC_{ij}$$

$$\therefore (G_C)_{ij} = \frac{(G\hat{Z})_{ij}}{D_i}$$

$$\begin{bmatrix} \frac{1}{D_1} \\ \frac{1}{D_2} \\ \vdots \\ \frac{1}{D_T} \end{bmatrix}_{Tx1} [1 \ 1 \ \dots \ 1]_{1 \times d} = \begin{bmatrix} \frac{1}{D_1} & \frac{1}{D_1} & \dots & \frac{1}{D_1} \\ \frac{1}{D_2} & \frac{1}{D_2} & \dots & \frac{1}{D_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{D_T} & \frac{1}{D_T} & \dots & \frac{1}{D_T} \end{bmatrix}$$

1_d^T
 $1 \times d$

$$\therefore \boxed{G_C^{(1)} = G\hat{Z} \odot (D^{-1} 1_d^T)} \quad G_C \text{ for path 1}$$

$T \times d$ $T \times d$ $T \times 1$ $1 \times d$

Path 2

$$d\mathcal{L} = \sum_{ij} (G\hat{Z})_{ij} dZ_{ij} = \sum_{ij} (G\hat{Z})_{ij} \left(-\frac{C_{ij}}{D_i^2}\right) dD_i = \sum_{ij} (G\hat{Z})_{ij} \frac{-\hat{Z}_{ij}}{D_i} dD_i$$

$$= \sum_{i=1}^T -\frac{1}{D_i} \left[\sum_{j=1}^d (G\hat{Z})_{ij} \hat{Z}_{ij} \right] dD_i$$

$$\therefore (G_D)_i = -\frac{1}{D_i} \sum_{j=1}^d (G\hat{Z})_{ij} \hat{Z}_{ij} \rightarrow \begin{bmatrix} G_D = D^{-1} \odot [(G\hat{Z} \odot \hat{Z}) 1_d] \end{bmatrix}$$

↑
row wise sum

$$\begin{bmatrix} G_D = D^{-1} \odot [(G\hat{Z} \odot \hat{Z}) 1_d] \end{bmatrix}$$

$T \times 1$ $T \times 1$ $T \times d$ $T \times d$ $d \times 1$

row wise sum

or

$$d\mathcal{L} = \sum_{ij} (G\hat{Z})_{ij} dZ_{ij} = \sum_{ij} (G\hat{Z})_{ij} -\frac{C_{ij}}{D_i^2} dD_i = \sum_{i=1}^T \sum_{j=1}^d -(G\hat{Z})_{ij} C_{ij} D_i^{-2} dD_i$$

$$\therefore (G_D)_i = -\sum_{j=1}^d (G\hat{Z})_{ij} C_{ij} D_i^{-2}$$

$$\boxed{G_D = -[(G\hat{Z} \odot C) 1_d] \odot D^{-2}}$$

$T \times 1$ $T \times d$ $T \times d$ $d \times 1$ $T \times 1$

$$D = \sqrt{G^2 + \epsilon} \quad \text{Back propagation}$$

Tx1 Tx1 Tx1

$$D_i = \sqrt{G_i^2 + \epsilon}$$

$$dD_i = \frac{1}{2\sqrt{G_i^2 + \epsilon}} dG_i^2 = \frac{1}{2D_i} dG_i^2$$

$$dL = \sum_{\lambda=1}^I (G_D)_i dD_i = \sum_{\lambda=1}^I (G_D)_i \frac{1}{2D_i} dG_i^2$$

$$\therefore (G_{G^2})_i = \frac{(G_D)_i}{2D_i} \rightarrow \boxed{G_{G^2} = \frac{1}{2} D^T \odot G_D}$$

Tx1 Tx1 Tx1

$$G^2 = \frac{1}{d} (C \odot C) 1_d \quad \text{Back propagation}$$

Tx1 Tx1 Txd Txd dx1

Tx1

row wise sum

$$G_i^2 = \frac{1}{d} \sum_{j=1}^d C_{ij}^2$$

$$dG_i^2 = \frac{1}{d} \sum_{j=1}^d dC_{ij}^2 = \frac{1}{d} \sum_{j=1}^d 2C_{ij} dC_{ij} = \sum_{j=1}^d \frac{2}{d} C_{ij} dC_{ij}$$

$$dL = \sum_{\lambda=1}^I (G_{G^2})_i dG_i^2 = \sum_{\lambda=1}^I (G_{G^2})_i \sum_{j=1}^d \frac{2}{d} C_{ij} dC_{ij}$$

$$= \sum_{ij} (G_{G^2})_i \frac{2}{d} C_{ij} dC_{ij}$$

$$\therefore (G_C^{(2)})_{ij} = \frac{2}{d} (G_{G^2})_i C_{ij}$$

Tx1 1xd Txd

$$\boxed{G_C^{(2)} = \frac{2}{d} (G_{G^2} 1_d^T) \odot C}$$

Txd

row wise broadcast

$$G_C^{(2)} = \frac{2}{d} [(\frac{1}{2} D^T \odot G_D) 1_d^T] \odot C = \frac{1}{d} [D^T \odot G_D] 1_d^T \odot C$$

$$\therefore \boxed{G_C = G_C^{(1)} + G_C^{(2)} = G_{G^2} \odot (D^T 1_d^T) + \frac{1}{d} [(D^T \odot G_D) 1_d^T] \odot C}$$

txd or Txd

$$\boxed{G_C = G_{G^2} \odot (D^T 1_d^T) + \frac{2}{d} (G_{G^2} 1_d^T) \odot C}$$

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$$C = Z - \mu \mathbf{1}_d^T \quad \text{Backpropagation}$$

$T \times d \quad T \times d \quad T \times 1 \quad 1 \times d$

$$C_{ij} = Z_{ij} - \mu_i$$

$$dC_{ij} = dZ_{ij} - d\mu_i \quad \text{* Note that } \mu \text{ is a function of } Z$$

therefore do not regard a constant.

$$d\mathcal{L} = \sum_{ij} (G_c)_{ij} dC_{ij}$$

$$= \sum_{ij} (G_c)_{ij} (dZ_{ij} - d\mu_i) = \sum_{ij} (G_c)_{ij} dZ_{ij} - \sum_{ij} (G_c)_{ij} d\mu_i$$

$$= \sum_{ij} (G_c)_{ij} dZ_{ij} - \sum_i \left(\sum_j (G_c)_{ij} \right) d\mu_i$$

$$\mu = \frac{1}{d} Z \mathbf{1}_d$$

↑
row wise sum

$$= \sum_{ij} (G_c)_{ij} dZ_{ij} - \sum_{ijk} \frac{1}{d} (G_c)_{ij} dZ_{ik}$$

substitute

$$= \sum_{ik} \left[(G_c)_{ik} - \sum_{j=1}^d \frac{1}{d} (G_c)_{ij} \right] dZ_{ik}$$

$$\mu_i = \frac{1}{d} \sum_{k=1}^d Z_{ik}$$

$$d\mu_i = \frac{1}{d} \sum_{k=1}^d dZ_{ik}$$

$$\therefore (G_z)_{ik} = (G_c)_{ik} - \frac{1}{d} \sum_{j=1}^d (G_c)_{ij}$$

$T \times d \quad T \times d$

↑
row wise sum
 $T \times d \times 1$

$$= (G_c)_{ik} - \frac{1}{d} (G_c \mathbf{1}_d)_i$$

$T \times d$

↑
need row wise broadcast $\mathbf{1}_d^T$

$$\therefore \boxed{G_z = G_c - \frac{1}{d} (G_c \mathbf{1}_d) \mathbf{1}_d^T}$$

$T \times d$

Layer Norm Backpropagation ends!

Compact LayerNorm Backpropagation

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Following is the derivation of compact LayerNorm Backward formula, which is mathematically equivalent to the original derivation but simpler to implement.

$$G_c^{(1)} = G_z \odot D^{-1} 1_d^T$$

$$G_c^{(2)} = \frac{\partial}{\partial C} (G_c \odot 1_d^T) \odot C \quad G_c^{(2)} = \frac{1}{2} D^{-1} \odot G_D$$

$$= \frac{\partial}{\partial C} \left[\left(G_D \odot \frac{1}{2D} \right) 1_d^T \right] \odot C \quad G_D = -[(G_z \odot C) 1_d] \odot D^{-2}$$

$$= \frac{\partial}{\partial C} \left[\left(-[(G_z \odot C) 1_d] \odot D^{-2} \right) \odot \frac{1}{2D} \right] 1_d^T \odot C$$

$$= -\frac{1}{d} \left[\left((G_z \odot C) 1_d \right) \odot D^{-3} \right] 1_d^T \odot C \quad C = \hat{Z} \odot D 1_d^T$$

where $(G_z \odot C) 1_d = (G_z \odot \hat{Z} \odot D 1_d^T) 1_d$

$$= D \odot [(G_z \odot \hat{Z}) 1_d]$$

$$= -\frac{1}{d} \left[D \odot [(G_z \odot \hat{Z}) 1_d] \right] \odot D^{-3} 1_d^T \odot C$$

$(A \odot (B 1_d^T)) 1_d = B \odot A 1_d$

$$= -\frac{1}{d} \left[D \odot [(G_z \odot \hat{Z}) 1_d] \right] \odot D^{-3} 1_d^T \odot C$$

$$= -\frac{1}{d} \left\{ D^{-2} \odot [(G_z \odot \hat{Z}) 1_d] \right\} 1_d^T \odot C \quad C = \hat{Z} \odot D 1_d^T$$

$$= -\frac{1}{d} \left\{ D^{-2} \odot [(G_z \odot \hat{Z}) 1_d] \right\} 1_d^T \odot (\hat{Z} \odot D 1_d^T)$$

$$= -\frac{1}{d} \left\{ D^{-2} 1_d^T \odot [(G_z \odot \hat{Z}) 1_d 1_d^T] \right\} \odot \hat{Z} \odot D 1_d^T$$

$$= -\frac{1}{d} \left\{ D^{-1} 1_d^T \odot [(G_z \odot \hat{Z}) 1_d 1_d^T] \right\} \odot \hat{Z}$$

$$= -\frac{1}{d} \left\{ D^{-1} \odot (G_z \odot \hat{Z}) 1_d \right\} 1_d^T \odot \hat{Z} = -\frac{1}{d} \left\{ D^{-1} \odot (G_z \odot \hat{Z}) \right\} 1_d^T \odot \hat{Z}$$

$$= -\frac{1}{d} \left[D^{-1} 1_d^T \odot (G_z \odot \hat{Z}) 1_d 1_d^T \right] \odot \hat{Z}$$

(13)

$$= -D^{-1}1_d^T \odot \frac{1}{d}(G_{\hat{z}} \odot \hat{z})1_d1_d^T \odot \hat{z}$$

mean

$$= -(D^{-1}1_d^T) \odot [\text{mean}(G_{\hat{z}} \odot \hat{z})1_d^T] \odot \hat{z}$$

$$\therefore G_c = G_c^{(1)} + G_c^{(2)}$$

$$= G_{\hat{z}} \odot D^{-1}1_d^T - (D^{-1}1_d^T) \odot [\text{mean}(G_{\hat{z}} \odot \hat{z})1_d^T] \odot \hat{z}$$

$$\boxed{G_c = (D^{-1}1_d^T) \odot [G_{\hat{z}} - \{\text{mean}(G_{\hat{z}} \odot \hat{z})1_d^T\} \odot \hat{z}]}$$

$\begin{matrix} T \times d & T \times 1 & 1 \times d & T \times d & T \times 1 & T \times d & T \times d & 1 \times d & T \times d \\ & & \text{Tx d} & & & & & & \\ & & & & T \times d & & & & \end{matrix}$

$$C = Z - \mu 1_d 1_d^T$$

$$C1_d = (Z - \mu 1_d 1_d^T)1_d = Z1_d - \mu \underbrace{1_d^T 1_d}_d = Z1_d - \mu d$$

$\mu = \frac{1}{d} Z1_d$

$$= Z1_d - \frac{1}{d} Z1_d d = Z1_d - Z1_d = 0$$

$$\hat{z} = C \odot D^{-1}1_d^T$$

$$\hat{z}1_d = C1_d \odot D^{-1}1_d^T 1_d = 0$$

$$G_c 1_d = [(D^{-1}1_d^T) \odot G_{\hat{z}}]1_d - \{ (D^{-1}1_d^T) \odot [\text{mean}(G_{\hat{z}} \odot \hat{z})1_d^T] \odot \hat{z} \} 1_d$$

$$G_c 1_d = D^{-1} \odot G_{\hat{z}} 1_d \quad [A \odot (B1_d^T)]1_d = B \odot A1_d$$

$$= d D^{-1} \odot \frac{1}{d} G_{\hat{z}} 1_d$$

$$G_c 1_d = d D^{-1} \odot \text{mean}(G_{\hat{z}})$$

$$G_z = G_c - \frac{1}{d}(G_c 1_d)1_d^T = G_c - \frac{1}{d}[d D^{-1} \odot \text{mean}(G_{\hat{z}})]1_d^T$$

$$= G_c - (D^{-1}1_d^T) \odot [\text{mean}(G_{\hat{z}})1_d^T]$$

$$\therefore \boxed{G_z = (D^{-1}1_d^T) \odot [G_{\hat{z}} - \{\text{mean}(G_{\hat{z}})1_d^T\} - \{\text{mean}(G_{\hat{z}} \odot \hat{z})1_d^T\} \odot \hat{z}]}$$

(14)

$$G_{\hat{z}} = G_x^{(w)} \odot \Gamma$$

$$\therefore G_z = \underset{\text{Tx} \times \text{d}}{(D^{-1} \mathbf{1}_d^T)} \odot \left[\underset{\text{Tx} \times \text{d}}{G_x^{(w)}} \odot \underset{\text{Tx} \times \text{d}}{\Gamma} - \underset{\text{Tx} \times \text{d}}{\text{mean}} \left(\underset{\text{Tx} \times \text{d}}{G_x^{(w)}} \odot \underset{\text{Tx} \times \text{d}}{\Gamma} \right) \underset{\text{Tx} \times \text{d}}{\mathbf{1}_d^T} \right. \\ \left. - \underset{\text{Tx} \times \text{d}}{\hat{z}} \odot \left\{ \underset{\text{Tx} \times \text{d}}{\text{mean}} \left[\underset{\text{Tx} \times \text{d}}{G_x^{(w)}} \odot \underset{\text{Tx} \times \text{d}}{\Gamma} \right] \odot \underset{\text{Tx} \times \text{d}}{\hat{z}} \right\} \underset{\text{Tx} \times \text{d}}{\mathbf{1}_d^T} \right]$$

Compact LayerNorm Backpropagation ends!

$Z = Y + F$ Backpropagation

This is very simple if trace is used and is skipped.

$$\boxed{\underset{T \times d}{G_Y} = \underset{T \times d}{G_Z}}$$

$$\boxed{\underset{T \times d}{G_F} = \underset{T \times d}{G_Z}}$$

$F = G W_2 + 1 + b_2$ Backpropagation

$$\boxed{\underset{T \times d_{ff}}{G_G} = \underset{T \times d \times d_{ff}}{G_F W_2^T}}$$

$$\boxed{\underset{d_{ff} \times d}{G W_2} = \underset{d_{ff} \times T}{G^T} \underset{T \times d}{G_F}}$$

$$\boxed{\underset{1 \times d}{G_{b_2}} = \underset{1 \times T}{1^T} \underset{T \times d}{G_F}}$$

$G = GELU(U)$ Backpropagation

$$G = GELU(U) = U \Phi(U)$$

$$G_{ij} = U_{ij} \Phi(U_{ij})$$

$$dG_{ij} = dU_{ij} \Phi(U_{ij}) + U_{ij} d\Phi(U_{ij}) \quad \frac{d\Phi(U_{ij})}{dU_{ij}} = \phi(U_{ij})$$

$$d\Phi(U_{ij}) = \phi(U_{ij}) dU_{ij}$$

$$= \Phi(U_{ij}) dU_{ij} + U_{ij} \phi(U_{ij}) dU_{ij}$$

$$= [\Phi(U_{ij}) + U_{ij} \phi(U_{ij})] dU_{ij}$$

$$dL = \sum_{i,j} (G_G)_{ij} dG_{ij} = \sum_{i,j} (G_G)_{ij} [\Phi(U_{ij}) + U_{ij} \phi(U_{ij})] dU_{ij}$$

$$\therefore G_{U_{ij}} = (G_G)_{ij} [\Phi(U_{ij}) + U_{ij} \phi(U_{ij})]$$

$$\rightarrow \boxed{\underset{T \times d_{ff}}{G_U} = \underset{T \times d_{ff}}{G_G} \odot [\underset{T \times d_{ff}}{\Phi(U)} + \underset{T \times d_{ff}}{U} \odot \underset{T \times d_{ff}}{\phi(U)}]}$$

$$G = \text{GELU}(x) \approx \frac{x}{2} [1 + \tanh(\sqrt{\frac{2}{\pi}}(x + 0.044715x^3))]]$$

Backpropagation

substitute $U = \sqrt{\frac{2}{\pi}}(x + 0.044715x^3)$

$$G = \frac{x}{2} (1 + \tanh U) = \frac{x}{2} \odot [1 + \tanh U]$$

Direct path: $dG = \frac{1}{2} dx \odot (1 + \tanh U)$

$$dL = \text{tr}(G_G^T dG)$$

$$= \text{tr}[G_G^T (\frac{1}{2} dx \odot (1 + \tanh U))]]$$

$$= \text{tr}[(G_G \odot (1 + \tanh U))^T \frac{1}{2} dx] \quad \text{tr}[A^T(B \odot C)] = \text{tr}[(A \odot C)^T B]$$

$$= \text{tr}[(G_G \odot \frac{1}{2} (1 + \tanh U)) dx]$$

$$dL = \text{tr}[G_X^T dx]$$

$$\therefore G_X^{(1)} = G_G \odot \frac{1}{2} (1 + \tanh U)$$

Indirect path: $dG = \cancel{\frac{1}{2} dx \odot (1 + \tanh U)} + \frac{x}{2} \odot d(1 + \tanh U)$

$$dG = \frac{x}{2} \odot d(\tanh U)$$

$$d(A \odot B) = dA \odot B + A \odot dB$$

$$= \frac{x}{2} \odot (1 - \tanh^2 U) \odot dU$$

$$dL = \text{tr}(G_U^T dG)$$

$$= \text{tr}[G_U^T (\frac{x}{2} \odot (1 - \tanh^2 U) \odot dU)]$$

$$= \text{tr}[(G_U \odot (\frac{x}{2} \odot (1 - \tanh^2 U))]^T dU] \quad \text{tr}[A^T(B \odot C)] = \text{tr}[(A \odot B)^T C]$$

$$\therefore G_U = G_G \odot \frac{x}{2} \odot (1 - \tanh^2 U)$$

$$U = \sqrt{\frac{2}{\pi}}(x + 0.044715x^3)$$

$$\frac{dU}{dx} = \sqrt{\frac{2}{\pi}}(1 + 0.134145x^2)$$

$$dU = \sqrt{\frac{2}{\pi}}(1 + 0.134145x^2) \odot dx$$

$$dL = \text{tr}(G_U^T dU)$$

$$= \text{tr}[G_U^T \{ \sqrt{\frac{2}{\lambda}} (1 + 0.134145 X^2) \odot dx \}]$$

$$= \text{tr}[\{G_U \odot \sqrt{\frac{2}{\lambda}} (1 + 0.134145 X^2)\}^T dx]$$

$$dL = \text{tr}(G_X^{(2)} dx)$$

$$\therefore G_X^{(2)} = G_U \odot \sqrt{\frac{2}{\lambda}} (1 + 0.134145 X^2)$$

$$G_X = G_X^{(1)} + G_X^{(2)} = G_G \odot \frac{1}{2} (1 + \tanh U) + G_U \odot \sqrt{\frac{2}{\lambda}} (1 + 0.134145 X^2)$$

$$= G_G \odot \frac{1}{2} (1 + \tanh U)$$

$$+ G_G \odot \frac{X}{2} \odot (1 - \tanh U) \odot \sqrt{\frac{2}{\lambda}} (1 + 0.134145 X^2)$$

$$G_X = G_G \odot \left[\frac{1}{2} (1 + \tanh U) + \frac{X}{2} (1 - \tanh U) \sqrt{\frac{2}{\lambda}} (1 + 0.134145 X^2) \right]$$

$$U = YW_i + 1_T b_i \quad \text{Backpropagation}$$

(16)

$$\boxed{G_{Y_{\text{FED}}} = G_U W_i^T} \quad \boxed{G_{W_i} = Y^T G_U} \quad \boxed{G_{b_i} = 1_T^T G_U}$$

$\text{Txd} \quad \text{Txdff} \quad \text{dffxd} \quad \text{dxTff} \quad \text{dxT} \quad \text{Txdff} \quad \text{1xdff} \quad \text{1xT} \quad \text{Txdff}$

$G_Y = G_{Y.\text{direct}} + G_{Y.\text{FED}} \quad \therefore G_Y = G_Z + G_U W_i^T$
 The next Backpropagation is $Y = \text{LayerNorm}(R)$ and
 the operation is exactly same as described earlier.

$$\boxed{G_R = G_C - \frac{1}{d} (G_C 1_d) 1_d^T}$$

$\text{Txd} \quad \text{Txd} \quad \text{Txd} \quad \text{dx1} \quad \text{1xd}$

$$\boxed{R^{(N)} = X^{(N-1)} + O} \quad \text{Backpropagation}$$

$$\boxed{G_{X_{\text{direct}}^{(N-1)}} = G_{TR}} \quad \boxed{G_O = G_{TR}}$$

$\text{Txd} \quad \text{Txd} \quad \text{Txd} \quad \text{Txd}$

$$O = \text{Concat}(O_1, O_2, \dots, O_H) W_0$$

$$\boxed{G_{\text{Concat}} = G_O W_0^T} \quad \boxed{G_{W_0} = \text{Concat}^T G_O}$$

$\text{Txd} \quad \text{Txd} \quad \text{dxd} \quad \text{dxd} \quad \text{dxt} \quad \text{Txd}$

$$G_{O_h} = G_{\text{Concat}}[:, (h-1)d_h : hd_h]$$

$$O_h = A_h V_h \quad \text{Backpropagation}$$

$$\boxed{G_{A_h} = G_{O_h} V_h^T} \quad \boxed{G_{V_h} = A_h^T G_{O_h}}$$

$\text{Txt} \quad \text{Txdh} \quad \text{dhxt} \quad \text{Txdh} \quad \text{Txt} \quad \text{Txdh}$

$A_h = \underset{\text{row}}{\text{softmax}}(\hat{S}_h)$ Backpropagation

(19)

$$\frac{\partial A_{h,ij}}{\partial \hat{S}_{h,ik}} = A_{h,ij} (\delta_{jk} - A_{h,ik}) \text{ where } \delta_{jk} = \begin{cases} 1 & j=k \\ 0 & j \neq k \end{cases}$$

$$dA_{h,ij} = \sum_{k=1}^T \frac{\partial A_{h,ij}}{\partial \hat{S}_{h,ik}} d\hat{S}_{h,ik} = \sum_{k=1}^T A_{h,ij} (\delta_{jk} - A_{h,ik}) d\hat{S}_{h,ik}$$

$$dL = \sum_{ij} G_{A_{h,ij}} dA_{h,ij} = \sum_{ijk} G_{A_{h,ij}} A_{h,ij} (\delta_{jk} - A_{h,ik}) d\hat{S}_{h,ik}$$

$$= \sum_{ik} \sum_{j=1}^T G_{A_{h,ij}} A_{h,ij} (\delta_{jk} - A_{h,ik}) d\hat{S}_{h,ik}$$

$$= \sum_{ik} G_{A_{h,ij}} \left(\sum_{j=1}^T A_{h,ij} \delta_{jk} - \sum_{j=1}^T A_{h,ij} A_{h,ik} \right) d\hat{S}_{h,ik}$$

row wise sum

$$\therefore G_{\hat{S}_{h,ik}} = \sum_{j=1}^T G_{A_{h,ij}} A_{h,ij} \delta_{jk} - \sum_{j=1}^T G_{A_{h,ij}} A_{h,ij} A_{h,ik}$$

$$= G_{A_{h,ik}} A_{h,ik} - A_{h,ik} \sum_{j=1}^T G_{A_{h,ij}} A_{h,ij}$$

row wise sum 1_T

$$G_{\hat{S}_h} = (G_{A_h} \odot A_h) - A_h \odot \left(\left[(G_{A_h} \odot A_h) 1_T \right] 1_T^T \right)$$

row wise broadcasting

$$\boxed{G_{\hat{S}_h} = A_h \odot \left[G_{A_h} - ((G_{A_h} \odot A_h) 1_T) 1_T^T \right]}$$

TXT TXT TXT TXT TXT TXI IXT

$$\tilde{S}_h = S_h + M \quad \text{Backpropagation}$$

(18)

$$\boxed{G_{S_h} = G_{\tilde{S}_h}}$$

TxT TxT

M is constant. Therefore $dM = 0$

No need to consider for Backpropagation.

$$S_h = \frac{Q_h K_h^T}{\sqrt{d_h}} \quad \text{Backpropagation}$$

$$dS_h = \frac{1}{\sqrt{d_h}} (dQ_h K_h^T + Q_h dK_h^T)$$

$$d\mathcal{L} = \text{tr}(G_{S_h}^T dS_h) = \frac{1}{\sqrt{d_h}} \text{tr}[G_{S_h}^T (dQ_h K_h^T + Q_h dK_h^T)]$$

$$= \frac{1}{\sqrt{d_h}} \text{tr}(G_{S_h}^T dQ_h K_h^T) + \frac{1}{\sqrt{d_h}} \text{tr}(G_{S_h}^T Q_h dK_h^T)$$

$$= \frac{1}{\sqrt{d_h}} \text{tr}(K_h^T G_{S_h}^T dQ_h) + \frac{1}{\sqrt{d_h}} \text{tr}(dK_h^T G_{S_h}^T Q_h)$$

$$= \frac{1}{\sqrt{d_h}} \text{tr}[(G_{S_h} K_h)^T dQ_h] + \frac{1}{\sqrt{d_h}} \text{tr}[Q_h^T G_{S_h} dK_h]$$

$$\therefore G_{Q_h} = \frac{1}{\sqrt{d_h}} G_{S_h} K_h \quad \quad \quad G_{K_h} = \frac{1}{\sqrt{d_h}} G_{S_h}^T Q_h$$

$$\therefore \boxed{G_{Q_h} = \frac{1}{\sqrt{d_h}} G_{S_h} K_h}$$

Tx dh TxT Tx dh

$$\boxed{G_{K_h} = \frac{1}{\sqrt{d_h}} G_{S_h}^T Q_h}$$

Tx dh TxT Tx dh

(19)

$$Q_h = X^{(N-1)} W_Q^h \quad K_h = X^{(N-1)} W_K^h \quad \text{Backpropagation}$$

$$\boxed{\underset{\text{Txd}}{G_{X^{(N-1)}}^{Q_h}} = \underset{\text{Txdh}}{G_{Q_h}} \underset{\text{dnhxd}}{(W_Q^h)^T}}$$

$$\boxed{\underset{\text{dxdh}}{G_{W_Q^h}} = \underset{\text{dxt}}{(X^{(N-1)})^T} \underset{\text{Txdh}}{G_{Q_h}}}$$

$$\boxed{\underset{\text{Txd}}{G_{X^{(N-1)}}^{K_h}} = \underset{\text{Txdh}}{G_{K_h}} \underset{\text{dnhxd}}{(W_K^h)^T}}$$

$$\boxed{\underset{\text{dxdh}}{G_{W_K^h}} = \underset{\text{dxt}}{(X^{(N-1)})^T} \underset{\text{Txdh}}{G_{K_h}}}$$

$$V_h = X^{(N-1)} W_V^h \quad \text{Backpropagation}$$

$$\text{Use } G_{V_h} = A_h^T G_{O_h}$$

$$\boxed{\underset{\text{Txd}}{G_{X^{(N-1)}}^{V_h}} = \underset{\text{Txdh}}{G_{V_h}} \underset{\text{dnhxd}}{(W_V^h)^T}}$$

$$\boxed{\underset{\text{dxdh}}{G_{W_V^h}} = \underset{\text{dxt}}{(X^{(N-1)})^T} \underset{\text{Txdh}}{G_{V_h}}}$$

$$\therefore G_{X^{(N-1)}}^{\text{head}} = G_{X^{(N-1)}}^{Q_h} + G_{X^{(N-1)}}^{K_h} + G_{X^{(N-1)}}^{V_h}$$

$$\therefore \boxed{\underset{\text{Txd}}{G_{X^{(N-1)}}^{\text{head}}} = \underset{\text{Txd}}{G_{Q_h}} \underset{\text{Txd}}{(W_Q^h)^T} + \underset{\text{Txd}}{G_{K_h}} \underset{\text{Txd}}{(W_K^h)^T} + \underset{\text{Txd}}{G_{V_h}} \underset{\text{Txd}}{(W_V^h)^T}}$$

$$G_{X^{\text{attention}}} = \sum_{h=0}^{H-1} G_{X^{\text{head}}}^{(h)}$$

$$\boxed{G_{X^{(N-1)}} = G_{X^{\text{attention}}} + G_{X^{\text{direct}}}$$

$$G_{X^{\text{direct}}} =$$

Embedding E Backpropagation

$$G_{E^{(LM)}} = G_L^T X^{(N)}$$

$$G_{E^{embedding}} = G_E[j, :] = \sum_{i: \text{input_ids}[i] = j} G_X[i, :]$$

↑
all i such that $\text{input_ids}[i]$ equals j

$$G_E = G_{E^{(LM)}} + G_{E^{embedding}}$$

$$X^{(N+1)} = X + P$$

$$\boxed{\underset{\text{txd}}{G_X} = \underset{\text{txd}}{G_{X^{(N+1)}}}}$$

$$\boxed{\underset{\text{txd}}{G_P} = \underset{\text{txd}}{G_{X^{(N+1)}}}}$$

All Backpropagations are done!

Update using $W \leftarrow W - \eta G_W$